

Tax Evasion and Productivity

Hans Martinez

Western University

September 11, 2026

Recap

Last meeting: What is an interesting question to wrap up the JMP?

- ▶ Counterfactual question: Optimal tax rate

Today:

- ▶ SOC of firm's optimization problem
- ▶ Functional forms for detection probability and tax evasion cost
- ▶ Estimation strategy

Introduction

- ▶ What's the question?
- ▶ Why should you care?

Contribution

- ▶ What do I do?
- ▶ How do I do it?

Overview of Results

- ▶ What do I find?
- ▶ Why is it important?

Colombia

Model

The Model

$$\begin{aligned} \max_{K_{it}, L_{it}, M_{it}, e_{it}} \quad & (1 - q(e_{it})) \left[P_t \mathbb{E}_\varepsilon[Y_{it}] - \rho_t M_{it} - r_t K_{it} - w_t L_{it} \right. \\ & \left. - \tau (P_t \mathbb{E}_\varepsilon[Y_{it}] - \rho_t (M_{it} + e_{it})) \right] \\ & + q(e_{it}) \left[(1 - \tau) (P_t \mathbb{E}_\varepsilon[Y_{it}] - \rho_t M_{it}) - r_t K_{it} - w_t L_{it} \right. \\ & \left. - \phi \rho_t \tau e_{it} \right] - \kappa_{it}(e_{it}, \omega_{it}) \\ \text{s.t.} \quad & Y_{it} = K_{it}^{\alpha_K} L_{it}^{\alpha_L} M_{it}^\beta \exp(\omega_{it} + \varepsilon_{it}) \end{aligned} \tag{1}$$

Important Assumptions

- ▶ Following seminal work of Allingham and Sandmo (1972), tax authority audits firms at random and always detects tax evasion
 - ▶ The probability of detection, $q(e_{it})$, is a function of the level of tax evasion, e_{it} . It might depend on other exogenous variables such as the monitoring technology of the tax authority, for example.
 - ▶ Impose assumption from Empirical model which assumes that choosing M does not enter the probability of detection or the cost of evasion. In other words, choosing M is independent of choosing e.

Important Assumptions (Cont.)

The cost of evasion, $\kappa_{it}(e_{it}, \omega_{it})$: any resources of the firm allocated to the tax evasion activity that could have been employed in a productive activity (e.g. manager's time spent on tax evasion instead of managing the firm).

- ▶ Unlike the penalty of tax evasion, ϕ , it does not depend on whether you are detected or not
- ▶ Increasing in the level of tax evasion, e_{it}
- ▶ Decreasing in firm's productivity, ω_{it} , but spikes at the top of the productivity distribution.

Important Assumptions - discussion on κ

- ▶ More productive firms have more resources to allocate to tax evasion, but since huge firms have access to more sophisticated legal tax avoidance strategies, the cost of illegal tax evasion is higher for the most productive firms.
- ▶ It could also be the case that the probability of detection is flat with respect to productivity except for huge (top productive) firms which could have a probability close to 1.
- ▶ Empirically, without data on cost of evasion or auditing information, q and κ are not separately identified. During estimation, we assume functional forms that allow us to reproduce the empirical patterns of tax evasion across the productivity distribution.

Model (Cont.)

The firms' problem (Equation 1) can be rewritten as follows,

$$\begin{aligned} \max_{K_{it}, L_{it}, M_{it}, e_{it}} \quad & \mathbb{E}[\pi_{it} | \Gamma] = (1 - \tau) (P_t \mathbb{E}[Y_{it}] - \rho_t M_{it}) - r_t K_{it} - w_t L_{it} \\ & + [1 - q(e_{it})] (\rho_t \tau e_{it}) - q(e_{it}) \phi \rho_t \tau e_{it} - \kappa_{it}(e_{it}, \omega_{it}) \\ \text{s.t.} \quad & Y_{it} = K_{it}^{\alpha_K} L_{it}^{\alpha_L} M_{it}^{\beta} \exp(\omega_{it} + \varepsilon_{it}) \end{aligned} \tag{2}$$

First Order Condition (FOC)

An interior solution leads to following first-order conditions,

$$\beta K_{it}^{\alpha_K} L_{it}^{\alpha_L} M_{it}^{-(1-\beta)} \exp(\omega_{it}) = \frac{\rho_t}{P_t} \quad (3)$$

$$\tau \rho_t (1 - (1 + \phi) (q(e_{it}) + q'(e_{it})e_{it})) = \frac{\partial \kappa_{it}(e_{it}, \omega_{it})}{\partial e_{it}} \quad (4)$$

(ε_{ijt} is measurement error, so $\mathcal{E} \equiv E[\exp(\varepsilon_{ijt})] = 1$)

Tax Evasion

To better understand the incentives to overreport inputs, we can rewrite Equation 4 in terms of the evasion elasticity of the probability of detection, $\xi_{q,e} = \frac{\partial q(e_{it})}{\partial e_{it}} \frac{e_{it}}{q(e_{it})}$, as follows,

$$\tau \rho_t \left(1 - (1 + \phi) (1 + \xi_{q,e}) q(e_{it}) \right) = \frac{\partial \kappa_{it}(e_{it}, \omega_{it})}{\partial e_{it}} \quad (5)$$

Corner Solution

Note that some firms will optimally choose not to evade, $e_{it} = 0$.

For these firms, the marginal cost of tax evasion is greater than the marginal benefit for any non-zero level of tax evasion.

This implies that for these firms, Equation 4 becomes,

$$\tau \rho_t \leq \frac{\partial \kappa_{it}(0, \omega_{it})}{\partial e_{it}} \quad (6)$$

Second Order Condition (SOC)

To guarantee a unique solution, the hessian matrix of the firm's problem must be negative definite.

$$\nabla^2 \mathbb{E}[\pi_{it} | \Gamma] = \begin{bmatrix} \frac{\partial^2 \mathbb{E}[\pi_{it} | \Gamma]}{\partial e_{it}^2} & 0 & 0 & 0 \\ 0 & \frac{\partial^2 \mathbb{E}[\pi_{it} | \Gamma]}{\partial M_{it}^2} & \frac{\partial^2 \mathbb{E}[\pi_{it} | \Gamma]}{\partial M_{it} \partial L_{it}} & \frac{\partial^2 \mathbb{E}[\pi_{it} | \Gamma]}{\partial M_{it} \partial K_{it}} \\ 0 & \frac{\partial^2 \mathbb{E}[\pi_{it} | \Gamma]}{\partial L_{it} \partial M_{it}} & \frac{\partial^2 \mathbb{E}[\pi_{it} | \Gamma]}{\partial L_{it}^2} & \frac{\partial^2 \mathbb{E}[\pi_{it} | \Gamma]}{\partial L_{it} \partial K_{it}} \\ 0 & \frac{\partial^2 \mathbb{E}[\pi_{it} | \Gamma]}{\partial K_{it} \partial M_{it}} & \frac{\partial^2 \mathbb{E}[\pi_{it} | \Gamma]}{\partial K_{it} \partial L_{it}} & \frac{\partial^2 \mathbb{E}[\pi_{it} | \Gamma]}{\partial K_{it}^2} \end{bmatrix} \quad (7)$$

- ▶ For a negative definite hessian matrix, the leading principal minors must alternate signs,
 - ▶ odd order minors must be negative and even-order minors positive.
- ▶ In our case, since
 - ▶ 1) the production function is strictly concave (CD) and
 - ▶ 2) choosing the level of tax evasion does not depend on the production function inputs; second cross derivatives are zero,
 - ▶ we would only have to check the first leading principal minor —the second derivative with respect to e_{it} — is negative,

$$\frac{\partial^2 \mathbb{E}[\pi_{it} | \Gamma]}{\partial e_{it}^2} = -\tau \rho_t (1 + \phi) (2q'(e_{it}) + e_{it} q''(e_{it})) - \frac{\partial^2 \kappa_{it}(e_{it}, \omega_{it})}{\partial e_{it}^2} < 0 \quad (8)$$

For strict concavity, we need only one of the following two conditions to be strictly positive,

$$\frac{\partial^2 \kappa_{it}(e_{it}, \omega_{it})}{\partial e_{it}^2} \geq 0 \quad (9)$$

$$\tau \rho_t (1 + \phi) (2q'(e_{it}) + e_{it} q''(e_{it})) \geq 0 \quad (10)$$

The second condition can be simplified to

$$-\frac{q''(e_{it})}{q'(e_{it})}e_{it} < 2 \quad (11)$$

since economic theory implies that $q'(e_{it}) > 0$ and $e_{it} > 0$, this condition imposes a **restriction on the curvature** of the detection function, $q(e_{it})$

- ▶ **Linear** detection **always** satisfies the SOC, since $q''(e_{it}) = 0$.
- ▶ **Convex** also satisfies the SOC **always**, since $q''(e_{it}) > 0$.
- ▶ **Concave** works if not too concave, i.e. **inequality holds**.

What does this inequality imply in practice?

With $\phi = 0$, logs of the interior FOC $\tau\rho_t(1 - q - q'e) = \partial\kappa/\partial e$ define

- ▶ the **log marginal benefit** $h(e) \equiv \ln(\tau\rho_t) + \ln(1 - q(e) - q'(e)e)$,
- ▶ with FOC $h(e) = \ln(\partial\kappa/\partial e)$.

	$q(e)$	SOC	$h(e) - \ln(\tau\rho_t)$	FOC ceiling
Linear	$\min\{\lambda e, 1\}$	$0 < 2$	$\ln(1 - 2\lambda e)$	$\lambda e < \frac{1}{2}$
Concave	$1 - e^{-\lambda e}$	$\lambda e < 2$	$-\lambda e + \ln(1 - \lambda e)$	$\lambda e < 1$
Convex	$\min\{(\lambda e)^k, 1\}$ $k > 1$	$1 - k < 2$	$\ln(1 - (1+k)(\lambda e)^k)$	$\lambda e < (1+k)^{-1/k}$

- ▶ All satisfy the SOC on their interior (exp's $\lambda e < 2 \iff \lambda e < 1$);
- ▶ All impose a restriction on λe .

A detection technology: the audit threshold

- ▶ The firm chooses overreporting e .
- ▶ The tax authority draws a threshold $T \sim U[0, 1/\lambda]$ and audits the firm if $e \geq T$.
 - ▶ An audit always uncovers evasion (Allingham and Sandmo 1972).
 - ▶ T is random to the firm (audit selection, third-party cross-checks, which inspector).

$$q(e) = \Pr(T \leq e) = \min\{\lambda e, 1\}.$$

- ▶ **Threshold scale:** $1/\lambda$ is the largest overreporting any audit draw tolerates; the *average* audit-triggering threshold is $1/2\lambda$.
- ▶ **Evasion bound:** the FOC stops firms at $\lambda e < \frac{1}{2}$
 - ▶ the most aggressive evader overreports below the *average* audit-triggering threshold,
 - ▶ at threshold, $q(1/2\lambda) = 50\%$ (a coin flip)

If $q(e)$ is concave, $T \sim \text{Exp}(\lambda)$ and $q(e) = 1 - e^{-\lambda e}$, mean threshold $1/\lambda$, detection $\approx 63\%$ at $e = 1/\lambda$.

How likely is the most aggressive evader caught?

At the FOC ceiling, the largest evasion drives the marginal benefit to zero,
 $q(b) + q'(b)b = 1$.

	b	$q(b)$	$\xi_{q,e} _{e=b}$	$E[T]$
Linear	$\frac{1}{2\lambda}$	50%	1 (unit elastic)	$\frac{1}{2\lambda}$
Exponential	$\frac{1}{\lambda}$	63%	$\frac{\lambda b}{e^{\lambda b} - 1} < 1$ (inelastic)	$\frac{1}{\lambda}$
Power	$\frac{1}{\lambda(1+k)^{1/k}}$	$\frac{1}{1+k} < 50\%$	$k > 1$ (elastic)	$\frac{k}{\lambda(k+1)}$

- ▶ $\xi_{q,e} = q'(e)e/q(e)$ is the evasion elasticity of the detection probability
- ▶ $\frac{1}{e-1} = 0.58$

Discussion

- ▶ Power (convex) form produces the lowest average audit-triggering threshold,
 - ▶ Exponential (concave) the largest.
 - ▶ Linear is intuitive: at max evasion, the tax authority flips a coin to audit/caught the firm
- ▶ An increase in 1% in the level of evasion increases the probability of being caught, by
 - ▶ 1% if linear (constant)
 - ▶ more than 1% if convex (constant)
 - ▶ less than 1% if concave (increasing), $\xi_{q,e}|_{e=0} = 0$ and $\xi_{q,e}|_{e=1/\lambda} = 0.58$.
- ▶ The extra dollar the firm hides increase its probability of detection more than the previous dollar.
- ▶ For linear and exponential, the FOC bounds evasion *at* the expected value of T
 - ▶ For power, FOC bounds evasion *below* its expected value: At $k = 2$, $\frac{1}{\lambda\sqrt{3}} < \frac{2}{\lambda 3}$

Estimation and recommendation

- ▶ **Linear** and **exponential** carry a single parameter λ
- ▶ **Power** $(\lambda e)^k$ adds a parameter, curvature k

Do **linear** and **power**,

- ▶ Start with **linear** ($k = 1$). Linear is the $k = 1$ case of the power form (ceiling $(1+k)^{-1/k} = \frac{1}{2}$).
- ▶ Then, estimate power (λ, k) , imposing feasibility $\lambda e < (1+k)^{-1/k}$ for *every* firm. Test if $k = 1$.

Empirical Strategies

- ▶ Testing for the presence of tax evasion by industry
 - ▶ Robust to different specifications
- ▶ Regression
 - ▶ Useful to study tax evasion heterogeneity across firms and over time
- ▶ Deconvolution
 - ▶ Recover distribution of tax evasion and productivity
 - ▶ Estimation of unbiased Production Function

Data

Testing

Recasting in M

The same trap reappears if we instead simulate $M_{it}^{(s)} = \beta P_t Y_{it} / \rho_t \cdot \mathbf{e}^{-\varepsilon^{(s)}}$ and form $e_{it}^{(s)} = M_{it}^* - M_{it}^{(s)}$ — this looks like it avoids $\mathcal{V}_{it}$, but

$$e(\varepsilon) = M_{it}^* (1 - \mathbf{e}^{-\mathcal{V}_{it} - \varepsilon}) = M_{it}^* - \underbrace{\frac{\beta P_t Y_{it}}{\rho_t} \mathbf{e}^{-\varepsilon}}_{= M(\varepsilon)} = M_{it}^* - M(\varepsilon)$$

using $\mathbf{e}^{-\mathcal{V}_{it}} = \beta P_t Y_{it} / (\rho_t M_{it}^*)$. **It's the identical map** — $\mathcal{V}_{it}$'s information is just M_{it}^* and $P_t Y_{it} / \rho_t$, so the M -version reconstructs it implicitly. Same nonlinearity, same missing reweighting, same problem.

What a valid simulated-moments design would need

- ▶ Match a **data-computable, ψ -free** population object against a forward-simulated analog — e.g. $E[\mathcal{V}_{it}|Z] = E[u_{it}|Z]$ is directly computable from data (since $\varepsilon \perp Z$), while $u^{(s)} = u(e^*(\psi^{(s)}, \omega^{(s)}; \theta))$ can be simulated forward from trial $(\psi^{(s)}, \omega^{(s)})$ draws
- ▶ This ties back to the deconvolved f_u from stage 1 as the natural target
- ▶ Delivers θ but not f_ψ , and needs its own machinery — no real gain over quadrature-MLE, which is already a cheap 1-D integral
- ▶ **Not pursued further** — MSL/quadrature is the headline strategy

An Alternative: ELVIS (Schennach (2014))

- ▶ The “Why not GMM” failure wasn’t fundamental — it was a *naive-averaging* bug: swapping in $\bar{\ell}_{it}(\lambda)$ silently drops the firm-specific reweighting
- ▶ Schennach (2014, *Econometrica*, “Entropic Latent Variable Integration via Simulation”) targets exactly this class: $E[g(U, Z, \theta)] = 0$ with U unobserved, g arbitrary — nonlinear, nonmonotone, nonseparable
- ▶ Her own canonical example (errors-in-variables: unobserved true regressor X^* , observed $X = X^* + \Delta X$) is structurally our M vs. M^*

Recasting the moment in M

Take M_{it} (true materials) — not ε_{it} — as the one unobservable. Everything else is a deterministic function of M and data:

$$e(M) = M_{it}^* - M, \quad \varepsilon(M) = \ln(M_{it}^*/M) - \mathcal{V}_{it}, \quad \omega(M) = \tilde{\mathcal{W}}_{it} - (1 - \beta)\varepsilon(M)$$

$$g(M, Z_{it}, \theta) \equiv \psi(M) = h(e(M)) - \delta_0 + \delta_1 \omega(M) - \delta_2 \omega(M)^2 \quad (12)$$

- ▶ Support $\mathcal{U}_{Z,\theta} = (M_{it}^* - \frac{1}{2\lambda}, M_{it}^*)$ — the feasibility ceiling $e < 1/(2\lambda)$ becomes the domain, not a complication (ELVIS explicitly allows θ -dependent support)
- ▶ Same ψ as before — recasting in M only changes *which variable ELVIS treats as unobserved*, not the moment itself

The entropic fix

Schennach's Theorem 2.1: the infinite-dimensional question “does *some* distribution of M given Z satisfy $E[g] = 0$ ” is **equivalent** to a finite-dimensional system in (θ, γ) :

$$\inf_{\mu} \|E_{\mu \times \pi}[g(M, Z, \theta)]\| = 0 \iff \inf_{\gamma \in \mathbb{R}^{d_g}} \|E_{\pi}[\tilde{g}(Z, \theta, \gamma)]\| = 0$$

$$\tilde{g}(Z, \theta, \gamma) \equiv \frac{\int g(M, Z, \theta) \exp(\gamma' g(M, Z, \theta)) d\rho(M|Z)}{\int \exp(\gamma' g(M, Z, \theta)) d\rho(M|Z)} \quad (13)$$

- ▶ The exponential tilt $\exp(\gamma' g(\cdot))$ **is** the missing reweighting — not assumed, but *solved for* as a Lagrange multiplier on the moment constraint
- ▶ $\rho(M|Z)$ is any user-chosen dominating measure — by her Remark 2.3, its choice has **zero effect** on the resulting estimate, only support matters

Estimation

Simulate $M^{(s)} \sim \rho(\cdot | Z_{it})$ (Metropolis) and form a ratio of averages:

$$\hat{g}(Z_{it}, \theta, \gamma) = \frac{\frac{1}{S} \sum_{s=1}^S g(M^{(s)}, Z_{it}, \theta) \exp(\gamma' g(M^{(s)}, Z_{it}, \theta))}{\frac{1}{S} \sum_{s=1}^S \exp(\gamma' g(M^{(s)}, Z_{it}, \theta))} \quad (14)$$

- ▶ Solve $(\hat{\theta}, \hat{\gamma})$ jointly by standard GMM on $E[\hat{g}(Z_{it}, \theta, \gamma)] = 0$
- ▶ λ identified the same way as before: sector–time variation in $\tau \rho_t$ shifting the evasion FOC's benefit
- ▶ 15 moments (finalized 2026-08-28), all built from independence assumptions the model already maintains — $\theta_{\text{smooth}} = (\delta_0, \delta_1, \delta_2, \sigma_\psi^2)$ (a new nuisance parameter, ψ 's own dispersion), λ gridded separately; full list in `Code/Rcpp/1200-stage2-elvis.cpp` / `CLAUDE.md`

ELVIS vs. MSL vs. plain MSM

	recovers f_ψ ?	needs f_ε ?	valid despite nonlinearity?
MSL (headline)	yes	yes	yes (by construction)
plain MSM	no	yes	no — naive-averaging bug
ELVIS	no	no — only support of M	yes (Thm 2.1)

- ▶ ELVIS's real payoff: stage 1's deconvolution of f_ε becomes **optional** for stage 2 — only $\hat{\beta}$ is needed to construct $\mathcal{V}, \tilde{\mathcal{W}}$
- ▶ $\hat{f}_\varepsilon$ can still be reused to build an efficient ρ (less simulation noise) — a computational choice, not an identification requirement

Going Further: One Unified Estimation

- ▶ If ELVIS works for stage 2 alone, it works for the *whole model* at once
- ▶ **There is only one true unobservable:** M_{it} — true materials, for every firm-period, corp and unincorporated alike
- ▶ No more $\mathcal{V}$, $\tilde{\mathcal{W}}$, no Jacobian, no density-transformation theorem — none of that machinery is needed once M itself is the latent
- ▶ Also sidesteps what made MSL “messier”: the AR(1) correction there needed the *conditional density* $f_{\omega_{it}|\omega_{it-1}}$ (i.e. f_{η} , recovered by Fourier deconvolution). ELVIS only needs **moment orthogonality** $E[\eta_{it}Z_{it}] = 0$ — no density of η at all

From M to ε

The materials FOC pins ε down exactly, given M :

$$\varepsilon_{it}(M_{it}) = \ln \beta - \ln \left(\frac{\rho_t M_{it}}{P_t Y_{it}} \right) \quad (15)$$

From M to ω — GNR's own trick, made exact

Substituting Equation 15 into the production identity

$\omega_{it} = y_{it} - \alpha_K k_{it} - \alpha_L l_{it} - \beta \ln M_{it} - \varepsilon_{it}$, the y_{it} terms **cancel exactly**:

$$\omega_{it}(M_{it}) = (1 - \beta) \ln M_{it} - \alpha_K k_{it} - \alpha_L l_{it} - \ln \beta + \ln \rho_t - \ln P_t \quad (16)$$

- ▶ This is **not information loss** — it's Gandhi-Navarro-Rivers's own $\Phi_{it} = y_{it} - \hat{\varepsilon}_{it}$ step, specialized to constant (Cobb-Douglas) elasticity: their Φ loses y_{it} the *same* way in this special case
- ▶ Standard proxy-variable logic (Olley-Pakes/Levinsohn-Petrin/ACF/GNR): under the timing assumption, optimal m is a monotonic function of ω *alone* — so m already perfectly proxies ω . y_{it} 's only job is revealing ε_{it} (via the share equation); once that's done, y_{it} has nothing further to say about ω
- ▶ y_{it} is used exactly **once** — inside $\varepsilon_{it}(M_{it})$

The AR(1) moment

Raw form (before substituting the FOC) — note the correct coefficients:

$$y_{it} - \beta \ln M_{it} - \alpha_K k_{it} - \alpha_L l_{it} - \gamma_0 - \gamma_1 (y_{it-1} - \beta \ln M_{it-1} - \alpha_K k_{it-1} - \alpha_L l_{it-1}) = \varepsilon_{it}$$

Since ε is itself just a function of M (Equation 15), substitute Equation 16 directly — no ε left anywhere:

$$\eta_{it}(M_{it}, M_{it-1}) = \omega_{it}(M_{it}) - \gamma_0 - \gamma_1 \omega_{it-1}(M_{it-1}) \quad (17)$$

Evasion moments, by firm type

Unincorporated (interior FOC), same as before:

$$e_{it}(M_{it}) = M_{it}^* - M_{it}, \quad \psi_{it}(M_{it}) = h(e_{it}(M_{it})) - \delta_0 + \delta_1 \omega_{it}(M_{it}) - \delta_2 \omega_{it}(M_{it})$$

Corp — relaxed from a hard constraint to a centering moment:

$$E[e_{it}(M_{it}) \mid \text{Corp}_i] = 0 \quad \text{or} \quad \text{Med}[e_{it}(M_{it}) \mid \text{Corp}_i] = 0$$

- ▶ No longer “corp never evade” imposed *outside* the estimation — corp can evade a little, just centered at zero, matching what the data already show (near-zero, imprecise evasion at both tails)

Unincorporated — evade on average, as an inequality:

$$E[e_{it}(M_{it}) \cdot \mathbb{1}(\text{Unincorp}_i) - U_2] = 0, \quad U_2 \geq 0$$

- ▶ ELVIS’s own device for inequalities (Example 1.3 in Schennach (2014)) — no separate technique needed

The full moment system

Moment	Object	Identifies	Instruments
Materials FOC AR(1) innovation	$\varepsilon_{it}(M_{it})$ $\eta_{it}(M_{it}, M_{it-1})$	β (mainly via corp) $\alpha_K, \alpha_L, \gamma_0, \gamma_1$	constant / sector k_{it}, l_{it} , lagged m^*
Evasion FOC (unincorp)	$\psi_{it}(M_{it})$	$\lambda, \delta_0, \delta_1, \delta_2$	sector-time $\times \tau \rho_t$
Corp evasion	$e_{it}(M_{it})$	pins $M_{it} \approx M_{it}^*$ for corp	— (centering)
Unincorp evasion sign	$e_{it}(M_{it}) \cdot$ $\mathbb{1}(\text{Unincorp})$	tightens the identified set	— (inequality)

- ▶ All firm-periods, corp and unincorporated, enter **jointly** — no data thrown away at any stage
- ▶ One draw of M_{it} (and its lag) feeds the *whole* moment vector — not double-counting, this is standard for any GMM system with multiple moments per observation

More moments from assumptions we already make

Every independence assumption already maintained — but never yet written as a moment — adds an (almost free) row to g :

Assumption	New moment
$\varepsilon \perp \psi$	$E[\varepsilon_{it}(M_{it})\psi_{it}(M_{it})] = 0$
$\omega \perp \varepsilon$	$E[\omega_{it}(M_{it})\varepsilon_{it}(M_{it})] = 0$
$\psi \perp \omega$	$E[\psi_{it}(M_{it})\omega_{it}(M_{it})] = 0$ — overidentifying check on the U-shape
$\varepsilon_{it} \perp \varepsilon_{it-1}$	$E[\varepsilon_{it}(M_{it})\varepsilon_{it-1}(M_{it-1})] = 0$ — free, M_{it-1} already drawn
$\varepsilon_{it-1} \perp \psi_{it}$	$E[\varepsilon_{it-1}(M_{it-1})\psi_{it}(M_{it})] = 0$ — free, same draw

- Formal license: Schennach (2014)'s own supplement, Prop. B.1 (conditional-mean-zero $\square$ countable unconditional moments $E[Y e^{i\xi X}] = 0$) and Prop. B.2 (independence, not just uncorrelatedness, factors the same way

What about the counterfactual?

MSL's revenue integral weights by $f_{\psi}(\psi(\varepsilon))J_i(\varepsilon)f_{\omega}(\omega(\varepsilon))f_{\varepsilon}(\varepsilon)$ — needs the full recovered density, which ELVIS doesn't deliver.

- ▶ ELVIS's own machinery for exactly this situation: Schennach's device for bounding a functional of the unobservable that isn't already one of the estimating moments — her worked example is literally recovering $E[U]$ this way (Schennach (2014), p.1250)
- ▶ Reusing the tilted density from the *main* estimation as if it were “the” posterior would give a number, but understate uncertainty — that tilted distribution is only the entropy-*maximizing* member of the set of distributions consistent with the moments, not necessarily the true one (she is explicit about this). The honest version re-solves for the functional's own identified set

Implementing it: an auxiliary moment

Hold $\hat{\theta}$ fixed (already estimated — the counterfactual is a forward simulation, not new estimation). Reuse the closed-form counterfactual evasion FOC already established:

$$\tilde{e}_i(\tilde{\tau}) = \tilde{e}(\tilde{\tau}; \omega_i(M_i), \psi_i(M_i)) = \frac{\tilde{\tau}\rho_t - C(\omega_i, \psi_i)}{2\lambda\tilde{\tau}\rho_t}, \quad C(\omega, \psi) \equiv \mathbf{e}^{\delta_0 - \delta_1\omega + \delta_2\omega^2}$$

(true M_i unchanged under the counterfactual — materials FOC doesn't depend on τ , per the model's own maintained assumptions) and form firm i 's counterfactual revenue as a function of the *same* latent M_i :

$$R_i(\tilde{\tau}; M_i) = \tilde{\tau} \left[P_t Y_{it} - \rho_t M_i - (1 - q(\tilde{e}_i(\tilde{\tau}))) \rho_t \tilde{e}_i(\tilde{\tau}) \right]$$

For a candidate $\tilde{\tau}$, introduce an auxiliary parameter $\tilde{\theta}(\tilde{\tau}) \equiv \frac{1}{N} \sum_i E[R_i(\tilde{\tau}) \mid Z_i]$ and **append its own moment** to the system:

$$E[R_i(\tilde{\tau}; M_i) - \tilde{\theta}] = 0$$

with θ fixed at $\hat{\theta}$ — only $\tilde{\theta}$ and an extended α (one more component, for the new

Point or set — decided by the procedure, not assumed

For a fixed $\tilde{\tau}$, grid over **candidate revenue values** $\tilde{\theta}$ (not over θ this time):

1. For each candidate $\tilde{\theta}$, solve the inner problem — maximize the objective (EL, or the CUE-type quadratic $-\frac{1}{2}\hat{g}'\hat{V}^{-1}\hat{g}$) over the extended γ , with the $R_i(\tilde{\tau}; M_i) - \tilde{\theta}$ moment included
2. That maximum is a criterion function $\hat{L}_n(\tilde{\theta}; \tilde{\tau})$ — exactly the same construction used for θ itself
3. **Profile against the unrestricted model:**
 $\hat{Q}_n(\tilde{\theta}; \tilde{\tau}) = -(\hat{L}_n(\tilde{\theta}; \tilde{\tau}) - \sup_{\theta', \gamma} \hat{L}_n(\theta', \gamma))$ — subtracting the global max, not using $\hat{L}_n$ raw, is what gives the efficiency gain this construction is built for
4. Region: $\{\tilde{\theta} : n\hat{Q}_n(\tilde{\theta}; \tilde{\tau}) \leq c_\alpha\}$. c_α via subsampling (Chernozhukov–Hong–Tamer 2007) for a tight region, or a fast conservative first pass: $c_\alpha = \chi^2_{d_g, \alpha}$ (table lookup, no subsampling — skips step 3's profiling)

► Repeat across a grid of $\tilde{\tau}$: **the shape decides**. A sharp trough at every $\tilde{\tau} \rightarrow$ the 42/97

Stage-2 ELVIS: Preliminary Results

Implementation

Moment set A (row list authoritative in `Code/Rcpp/1200-stage2-elvis.cpp`'s header comment): $\theta_{\text{smooth}} = (\lambda, \delta_0, \delta_1, \delta_2)$, 9 moments ($d_g = 9$), no nuisance parameters beyond γ :

$$g(M, Z, \theta_{\text{smooth}}) = \begin{pmatrix} \psi(M, Z) \\ \varepsilon(M, Z) \\ \psi(M, Z) \ln M \\ \psi(M, Z) \omega(M, Z) \\ \psi(M, Z) \omega(M, Z)^2 \\ \varepsilon(M, Z) \ln M \\ \varepsilon(M, Z) e(M, Z) \\ \varepsilon(M, Z) \omega(M, Z) \\ h'(e, \lambda) \varepsilon(M, Z) \end{pmatrix}$$

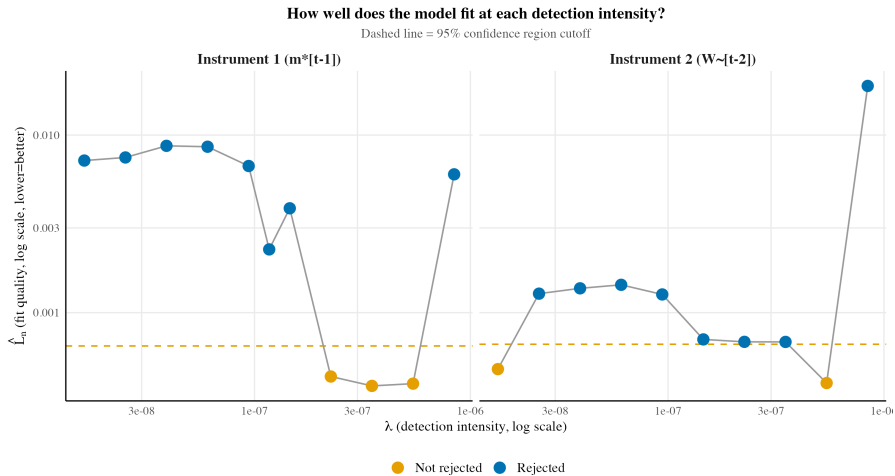
Implementation Details

- ▶ **Objective:** CUE quadratic, $\hat{L}_n(\theta, \gamma) = \frac{1}{2} \hat{g}' \hat{\Omega}^- \hat{g}$, $\hat{\Omega}^-$ an eigenvalue-truncated generalized inverse — same construction AK2020 use
- ▶ **Sampler:** per-firm Metropolis chain, `n_burn=1000`, `n_keep=3000`, deterministic per-firm RNG reseed
- ▶ **Solver:** Nelder-Mead (beats BOBYQA decisively on this problem), unbounded γ (a large/degenerate γ is an expected solution, Schennach 2014 — not a numerical failure), two-pass refinement
- ▶ **Seeding:** every λ grid point seeded independently from the same validated starting point — no chaining across grid points

λ Search

- ▶ λ bounds derived directly from the data: the smallest λ for which not even the largest firm's evasion is ever constrained, up to the largest λ for which even the smallest firm is constrained
- ▶ 16-point log-spaced grid (coarse pass, then a finer pass bracketing the region of interest)
- ▶ **Sample:** 32,232 unincorporated firm-periods (3,340 at the $\tau_P = 0$ corner), top 0.5% trimmed by M^* ; two productivity-instrument choices for $(\alpha_K, \alpha_L, \beta)$

$\hat{L}_n(\lambda)$ — Both Instruments



Reading the picture:

- ▶ m_{it-1}^* : a clean V-shape, genuine minimum, real rejection on both sides — a proper confidence interval
- ▶ $\tilde{\mathcal{W}}_{it-2}$: also rejects everywhere except two points — its own interior minimum, and separately the smallest λ we tested. That second pass just means: for this instrument, we can't rule out arbitrarily weak detection either — we only get an upper bound on λ from it, not a full interval. Consistent with it being the weaker of the two instruments overall.

Headline Estimates

Instru- ment	$\hat{\lambda}$	95% region	$\hat{L}_n$	$\hat{\delta}_0$	$\hat{\delta}_1$	$\hat{\delta}_2$	$\hat{\omega}^* = \hat{\delta}_1 / 2\hat{\delta}_2$	$\max \gamma $
m_{it-1}^*	3.50×10^{-7}	$[2.26, 5.43] \times 10^{-7}$	0.000387	3.459	4.315	0.541	3.99	0.64
$\tilde{\mathcal{W}}_{it-2}$	5.43×10^{-7}	5.43×10^{-7} (point)	0.000401	10.206	7.460	0.972	3.84	1.86

$\hat{\delta}_2 > 0$ for both — the cost function opens the correct (U-shaped) way. $\hat{\omega}^*$ lands in a similar plausible range for both, even though the individual $\hat{\delta}_1, \hat{\delta}_2$ differ — expected: the two instruments come from different first-stage residuals, so they carry somewhat different implied productivity distributions.

A Data-Side Check on $\hat{\lambda}$: What Detection Probability Does This Imply?

Does $\hat{\lambda}$ imply a sane detection probability at a typical evading firm? Quick check by forward-simulating each firm's own evasion at the fitted $(\hat{\theta}, \hat{\gamma})$ and reading off the population mean/median — **a ballpark reading, not the formal way to pin this down** (that would need its own auxiliary moment, jointly re-estimated with everything else, not a post-hoc average):

Instrument	$E[e]$	Med[e]	$q(E[e]) = \hat{\lambda} \cdot E[e]$	$q(\text{Med}[e])$
m_{it-1}^*	2,545	625	0.089%	0.022%
$\tilde{W}_{it-2}$	3,189	685	0.173%	0.037%

- ▶ $E[e] \gg \text{Med}[e]$ for both — a right-skewed evasion distribution: most firms evade modestly, a long right tail of aggressive evaders pulls the mean well above the median
- ▶ Both implied detection probabilities are small, well under the 50% FOC ceiling

Notes for Discussion

- ▶ m_{it-1}^* **is the instrument to lead with:** a genuine, contiguous confidence interval, stable $\hat{\delta}_1, \hat{\delta}_2$ across the whole search
- ▶ $\tilde{\mathcal{W}}_{it-2}$ **confirms the same qualitative story but is the weaker instrument:** real rejection pattern, a real interior optimum, but can only bound λ from above, not below
- ▶ **No formal joint confidence region for $(\lambda, \delta_1, \delta_2)$ yet** — next step, to get a stronger justification for one fixed $\hat{\theta}$ to carry into the counterfactual

Empirical Strategy

Functional Forms

The probability of detection, $q(e_{it})$, and the cost of evasion, $\kappa_{it}(e_{it}, \omega_{it})$, functions should display the following properties to be consistent with the observed patterns in the data and with theory:

- ▶ $q(e_{it}) = \min\{\lambda e, 1\}$
 - ▶ $q(e_{it}) \in [0, 1]$, $q(0) = 0$, $q(\infty) = 1$, and $q(e_{it})$ is increasing in e . Since I do not have variation in government audit effort, λ is treated as constant across firms and time
- ▶ $\kappa_{it}(e_{it}, \omega_{it}) = e_{it} \mathbf{e}^{\delta_0 - \delta_1 \omega_{it} + \delta_2 \omega_{it}^2 + \psi_{it}}$
 - ▶ $\kappa_{it}(e_{it}, \omega_{it}) \in [0, \infty)$, $\kappa_{it}(0, \omega_{it}) = 0$, $\frac{\partial \kappa_{it}(0, \omega_{it})}{\partial e_{it}} \geq 0$.
 - ▶ $\frac{\partial \kappa_{it}(e_{it}, \omega_{it})}{\partial \omega_{it}} \leq 0$ and $\frac{\partial^2 \kappa_{it}(e_{it}, \omega_{it})}{\partial \omega_{it}^2} \geq 0$ to be consistent with the observed patterns in the data.
 - ▶ U-shape not ideal, but simple and captures the idea of decreasing cost at high ω
 - ▶ Corporations out of the FOC
 - ▶ ψ_{it} idiosyncratic shock to the cost of evasion to explain tax evasion heterogeneity observed in empirical analysis

FOC

Using the functional forms defined above, the FOCs wrt tax evasion become for unincorporated firms,

$$\tau\rho_t(1 - 2\lambda e_{it}) = \mathbf{e}^{\delta_0 - \delta_1\omega_{it} + \delta_2\omega_{it}^2 + \psi_{it}}$$

Taking logs and defining

$$h(e_{it}) \equiv \ln(\tau\rho_t) + \ln(1 - 2\lambda e_{it}), \quad e_{it} \in (0, \frac{1}{2\lambda}),$$

the evasion FOC become

$$\psi_{it} = h(e_{it}) - \delta_0 + \delta_1\omega_{it} - \delta_2\omega_{it}^2 \tag{18}$$

How to estimate?

- ▶ $\tau\rho_t$ is *computed from data*

I do not observe e_{it} nor ω_{it} at the firm level.

- ▶ However, using the estimated **parameters of the production** function, we can
- ▶ rewrite the FOC in terms of data, parameters, and
- ▶ *only one* independent unobservable ε

How to estimate the production function parameters and the distribution of ε ? **Same as before**

Estimating β

Same as before: From Corporations, the truth-reporters, I recover

- ▶ the output elasticity of materials, β , and the
- ▶ distribution of the error term, f_ε ,

For Corporations, $M_{it}^* = M_{it}$, and thus,

$$\ln \left(\frac{\rho_t M_{it}^*}{P_t Y_{it}} \right) = \ln(\beta) - \varepsilon_{it}$$

Production Function Estimation

- Having recovered β , for everyone, I can form the following variable,

$$\begin{aligned}\mathcal{V}_{it} &\equiv \ln \left(\frac{\rho_t M_{it}^*}{P_t Y_{it}} \right) - \ln(\beta) \\ &= \ln \left(\frac{\rho_t M_{it}^*}{P_t Y_{it}} \right) - \ln \left(\frac{\rho_t M_{it}}{P_t Y_{it}} \right) + \ln \left(\frac{\rho_t M_{it}}{P_t Y_{it}} \right) - \ln(\beta) \\ &= \ln \left(\frac{M_{it}^*}{M_{it}} \right) - \varepsilon_{it}\end{aligned}\tag{19}$$

Production Function Estimation (Cont.)

- To recover the remaining production function parameters, the **original strategy is still valid**

To see why, note that,

$$\begin{aligned}\mathcal{W}_{it} &\equiv y_{it} - \beta(m_{it}^* - \mathcal{V}_{it}) \\ &= y_{it} - \beta(m_{it}^* - \ln\left(\frac{M_{it}^*}{M_{it}}\right) + \varepsilon_{it}) \\ &= y_{it} - \beta(m_{it}^* - \ln(M_{it}^*) + \ln(M_{it})) - \beta\varepsilon_{it} \\ &= y_{it} - \beta m_{it}^* + \beta \ln(M_{it}^*) - \beta \ln(M_{it}) - \beta\varepsilon_{it} \\ &= \alpha_K k_{it} + \alpha_L l_{it} + \omega_{it} + (1 - \beta)\varepsilon_{it}\end{aligned}\tag{20}$$

the last equality is *exactly the same* as the original discussed strategy

Using productivity's Markov process to recover remaining PF parameters

Therefore, the **previous approach is still valid** to recover the remaining production function parameters, productivity's Markov process, and the distribution of the productivity shock $(\alpha_K, \alpha_L, \omega_{it})$.

Assuming, productivity follows an AR(1) Markov process,

$$\omega_{it} = \gamma_0 + \gamma_1 \omega_{it-1} + \eta_{it} \quad (21)$$

we recover the PF parameters, by substituting Equation 20, into Equation 21,

$$\begin{aligned} \mathcal{W}_{it} = & \gamma_0 + \alpha_K k_{it} + \alpha_L l_{it} \\ & + \gamma_1 (\mathcal{W}_{it-1} - \alpha_K k_{it-1} - \alpha_L l_{it-1}) \\ & + (1 - \beta) \varepsilon_{it} - \gamma_1 (1 - \beta) \varepsilon_{it-1} + \eta_{it} \end{aligned} \quad (22)$$

Estimating Equation 22

By finding instruments, we can estimate Equation 22, using the conditional moment,

$$E[(1 - \beta)\varepsilon_{it} - \gamma_1(1 - \beta)\varepsilon_{it-1} + \eta_{it} | Z_{it}] = 0$$

- I use capital and labour as instruments for themselves and reported materials and $\mathcal{W}_{it-2}$ as instruments to control for endogeneity between $\mathcal{W}_{it-1}$ and

ε_{it-1}

- $Z = \{k_{it}, l_{it}, m_{it-1}^*, \mathcal{W}_{it-2}\}$

Using the estimated parameters of the production function

As previously discussed, with α_K and α_L at hand form,

$$\tilde{\mathcal{W}}_{it} \equiv \mathcal{W}_{it} - \alpha_K k_{it} - \alpha_L l_{it} = \omega_{it} + (1 - \beta)\varepsilon_{it} \quad (23)$$

and recover f_ω using deconvolution,

$$f_{\tilde{\mathcal{W}}} = \int f_\varepsilon \left(\frac{1}{1 - \beta} (\tilde{\mathcal{W}} - \omega) \right) f_\omega(\omega) d\omega$$

Would GMM work ?

Substituting and simplifying

$$\begin{aligned}\psi_{it} + \delta_1(1 - \beta)\varepsilon = & \ln(\tau\rho_t) + \ln(1 - 2\lambda M_{it}^*(1 - \mathbf{e}^{-\mathcal{V}_{it} - \varepsilon})) \\ & - \delta_0 + 2\delta_2(1 - \beta)\tilde{\mathcal{W}}_{it}\varepsilon - \delta_2(1 - \beta)^2\varepsilon^2 \\ & + \delta_1\tilde{\mathcal{W}}_{it} - \delta_2\tilde{\mathcal{W}}_{it}^2\end{aligned}\tag{24}$$

Taking conditional expectation

Using $\psi \perp Z$, $\varepsilon \perp Z$, and $\omega \perp \varepsilon$ (the same independence MSL already relies on).
 $\varepsilon \perp Z \Rightarrow E[\delta_1(1 - \beta)\varepsilon \mid Z] = \delta_1(1 - \beta)E[\varepsilon \mid Z] = 0$

$$\begin{aligned} 0 = & \ln(\tau\rho_t) + E[\ln(1 - 2\lambda M_{it}^*(1 - \mathbf{e}^{-\mathcal{V}_{it}-\varepsilon})) \mid Z] \\ & \underbrace{- \delta_0 - \delta_2(1 - \beta)^2\sigma_\varepsilon^2}_{\text{Constant (known from stage 1)}} \\ & + \delta_1\tilde{\mathcal{W}}_{it} - \delta_2\tilde{\mathcal{W}}_{it}^2 \end{aligned} \tag{25}$$

Why not GMM with instruments?

Everything in Equation 25 is a known function of data and θ , **except**:

$$\ell_{it}(\varepsilon; \lambda) \equiv \ln(1 - 2\lambda M_{it}^*(1 - \mathbf{e}^{-\mathcal{V}_{it} - \varepsilon}))$$

so the moment reads

$$0 = \ln(\tau \rho_t) + E[\ell_{it}(\varepsilon; \lambda) \mid Z] - \delta_0 - \delta_2(1 - \beta)^2 \sigma_\varepsilon^2 + \delta_1 \tilde{\mathcal{W}}_{it} - \delta_2 \tilde{\mathcal{W}}_{it}^2$$

Simulating the nonlinear term

If we hold firm it 's own $(M_{it}^*, \mathcal{V}_{it})$ fixed and simulate ε ,

$$\hat{\ell}_{it}^S(\lambda) = \frac{1}{S} \sum_{s=1}^S \ell_{it}(\varepsilon^{(s)}; \lambda), \quad \varepsilon^{(s)} \stackrel{iid}{\sim} \hat{f}_{\varepsilon}$$

By the LLN, as $S \rightarrow \infty$:

$$\hat{\ell}_{it}^S(\lambda) \xrightarrow{a.s.} \bar{\ell}_{it}(\lambda) \equiv \int \ell_{it}(\varepsilon; \lambda) f_{\varepsilon}(\varepsilon) d\varepsilon$$

A specific, computable number given $(M_{it}^*, \mathcal{V}_{it}, \lambda)$. Looks like it solved the problem — no more unobserved ε .

But it's the wrong number

The moment needs $\ell_{it}(\varepsilon_{it}; \lambda)$

— firm it 's value at its **own realized** ε_{it} — not the average over every value ε could have taken.

$\ell_{it}(\cdot; \lambda)$ is nonlinear in ε , so generically

$$\bar{\ell}_{it}(\lambda) \neq \ell_{it}(\varepsilon_{it}; \lambda)$$

Let $\eta_{it}(\lambda) \equiv \ell_{it}(\varepsilon_{it}; \lambda) - \bar{\ell}_{it}(\lambda) \neq 0$. Swapping in $\bar{\ell}_{it}(\lambda)$ silently injects $-\eta_{it}(\lambda)$ into every firm's moment.

Conclusion: this doesn't work

- ▶ $\eta_{it}(\lambda)$ is a nonlinear function of firm it 's own ε_{it} , entangled with $(M_{it}^*, \mathcal{V}_{it})$
- ▶ For Z to be *relevant* for λ , it must correlate with the firm's evasion behavior $(M_{it}^*, \mathcal{V}_{it})$, but $\eta_{it}(\lambda)$ depends on them
- ▶ $\varepsilon \perp Z$ alone doesn't deliver $E[Z \cdot \eta_{it}(\lambda_0)] = 0$
- ▶ Nonlinearity plus averaging over the unconditional marginal throws away exactly the firm-specific information tying ε_{it} to its own data

Implicit Assumptions

- ▶ The functional forms for $q(e)$ and $\kappa(e)$ are correctly specified and do not change over time
- ▶ No entry, no exit
- ▶ Corporations never evade
- ▶ Partial equilibrium, Inelastic demand, (No GE effects)
 - ▶ Tax increases do not change optimal input choices and output. FOC of materials does not depend on tax rate
 - ▶ Firms are already assumed to be price takers

Couterfactual Question

What Would Be The Optimal Tax Rate?

The government faces a trade-off:

- ▶ as the government increases the tax rate, the tax revenue increases
- ▶ as the tax rate increases, the marginal benefit of evading taxes increase leading to higher levels of tax evasion which reduces tax revenue

What is the optimal rate that maximizes expected tax revenue for the government?

What is the tax rate that lead to median (media) tax evasion of zero?

The Government's Problem

The government wants to choose the sales tax rate τ that maximizes the expected tax revenue from the sum of all firm i

$$\max_{\tau} \sum_i q(e) \tau [P_t Y_{ijt} - \rho_t M_{ijt}] + (1 - q(e)) \tau [P_t Y_{ijt} - \rho_t (M_{ijt} + e_{ijt})]$$

Simplifying,

$$\max_{\tau} \sum_i \tau [P_t Y_{ijt} - \rho_t M_{ijt} - (1 - q(e)) \rho_t e_{ijt}]$$

where the first two terms are the potential tax revenue (in the absence of tax evasion) and the last terms is the expected undetected tax evasion.

for firm i ,

$$\begin{aligned} & [P_t Y_{ijt} - \rho_t M_{ijt} - (1 - q(e)) \rho_t e_{ijt}] \\ & + \tau \left[-(1 - q(e)) \rho_t \frac{\partial e_{ijt}}{\partial \tau} + \rho_t e_{ijt} \frac{\partial q(e)}{\partial e} \frac{\partial e_{ijt}}{\partial \tau} \right] = 0 \end{aligned}$$

simplifying,

$$\begin{aligned} P_t Y_{ijt} - \rho_t M_{ijt} &= (1 - q(e)) \rho_t e_{ijt} \\ &+ \tau \rho_t \frac{\partial e_{ijt}}{\partial \tau} \left[(1 - q(e)) - \frac{\partial q(e)}{\partial e} e_{ijt} \right] \end{aligned}$$

Numerical Strategy

1. Having recovered $\theta = (\lambda, \delta_0, \delta_1, \delta_2)$ and f_ψ through MSL are held fixed for everything below
2. For each firm-year, recover the implied **type** (ω, ψ) at the *observed* τ , using the same map as the likelihood (Equation 36): for $M_{it}(\varepsilon) = M_{it}^* \mathbf{e}^{-\mathcal{V}_{it}-\varepsilon}$,

$$\omega(\varepsilon) = \tilde{\mathcal{W}}_{it} - (1 - \beta)\varepsilon, \quad \psi(\varepsilon) = h(e(\varepsilon)) - \delta_0 + \delta_1\omega(\varepsilon) - \delta_2\omega(\varepsilon)^2$$

3. Hold (ω, ψ) fixed as firm primitives (they don't respond to the tax rate) and resolve the evasion FOC at a proposed $\tilde{\tau}$ to find the counterfactual evasion, $\tilde{e}$. Under linear q , closed form, no root-finding:

$$\tilde{e}(\tilde{\tau}; \omega, \psi) = \frac{\tilde{\tau} \rho_t - C(\omega, \psi)}{2\lambda \tilde{\tau} \rho_t}, \quad C(\omega, \psi) \equiv \mathbf{e}^{\delta_0 - \delta_1 \omega + \delta_2 \omega^2 + \psi}$$

4. $\tilde{e}$ is a **deterministic function of the same** ε that generated (ω, ψ) . So expected revenue is a *single* integral over ε , weighted exactly as the likelihood weights it, normalized by the firm's own likelihood contribution $\mathcal{L}_i(\theta)$ (already computed in stage 2), the conditional expectation given firm i 's data:

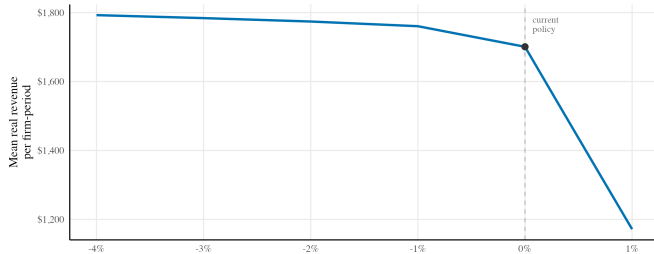
$$E[R_i(\tilde{\tau})] = \frac{1}{\mathcal{L}_i(\theta)} \int \tilde{\tau} \left[P_t Y_{it} - \rho_t M_{it}(\varepsilon) - (1 - q(\tilde{e}(\varepsilon))) \rho_t \tilde{e}(\varepsilon) \right] A(\varepsilon) d\varepsilon \quad (26)$$

where $A = f_\psi(\psi(\varepsilon)) J_i(\varepsilon) f_\omega(\omega(\varepsilon)) f_\varepsilon(\varepsilon)$

5. Evaluate Equation 26 with the same quadrature nodes used in stage 2
6. Find $\tilde{\tau}^* = \arg \max_{\tilde{\tau}} \sum_i E[R_i(\tilde{\tau})]$ by grid or golden-section search over the scalar $\tilde{\tau}$

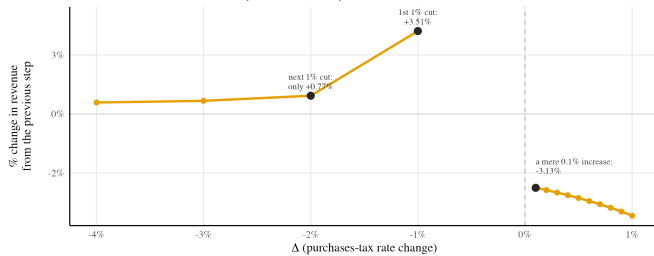
Revenue around current policy

Pure forward-simulation at the best-fit operating point — not yet run through the (ΔR) test-inversion grid



The headline in numbers

a 1% cut buys +3.51%, the next only +0.77% — a 0.1% increase alone costs -3.13%



Headline Result

At the current best-fit operating point (m_{it-1}^* , $\hat{\lambda} = 3.5 \times 10^{-7}$, $\hat{\delta}_1 = 4.3$, $\hat{\delta}_2 = 0.54$) — forward-simulated, not yet run through formal test inversion:

- ▶ A 1% cut in τ_P raises real revenue **+3.5%**
- ▶ The next 1% cut only buys **+0.8%** more — diminishing fast
- ▶ A mere 0.1% *increase* costs **-3.1%** — a tenth of the move undoes more than the first full point of cutting bought

Why

Linear q + linear $\kappa \Rightarrow$ new evasion is exactly linear in the firm's own baseline evasion, with an intercept identical across every firm:

$$\tilde{e}(\Delta) = \frac{e_i}{1 + \Delta} + \frac{\Delta}{2\lambda(1 + \Delta)}$$

$1/\lambda$ is large, so that intercept dominates almost immediately past $\Delta = 0$ — the same shift for every firm, regardless of size.

The Optimal Cut, Not the Corner

- ▶ Revenue keeps rising as $\Delta \rightarrow -1$ mathematically — no evasion left, government keeps 100% of the sales tax — but that's not a serious answer: ignores why a purchases-side credit exists at all (cascading taxation), and pushes past where (ω, ψ) can be taken as fixed
- ▶ Marginal revenue of the next cut flattens to a mechanical floor by $\Delta \approx -20\%$ to -25% — the smallest cut that already captures nearly all the behavioral benefit
- ▶ Not yet backed by a confidence set — point estimate only

Estimating Tax Evasion by Deconvolution

Objective

- ▶ Empirical exercise to motivate structural model

What can be deconvolved and what can't

Convolution (Meister, 2009)

The density of the sum of two **independent** random variables is equal to the **convolution** of the densities of both addends; hence

$$f_V = \int f_\varepsilon(V - u)f_u(u)du$$

Under $\varepsilon \perp\!\!\!\perp M, e$ and $e \perp\!\!\!\perp M$, with known f_ε and $u = \ln\left(\frac{M+e}{M}\right)$. Also $M^* = M + e$, and $V = u + \varepsilon$.

Then, I can deconvolve:

► $f_u: f_V = \int f_\varepsilon(V - u)f_u(u)du$

I don't need f_e , actually

What deconvolution recovers is the **overreporting log-ratio**,

$$u_{it} \equiv \ln \left(\frac{M_{it}^*}{M_{it}} \right) = \ln \left(1 + \frac{e_{it}}{M_{it}} \right),$$

- ▶ the *share of overreported over true inputs*
- ▶ the same object discussed previously, now given an explicit structural interpretation
- ▶ Since it's a ratio, it's adjusted by the firm's true materials level and hence informative about the intensity of tax evasion
- ▶ Had I recovered e , I would have to normalize somehow to compare firms of different levels of material utilization

Getting rid of the log

If I'm interested in only the density of the ratio, I can use a **density transformation**

Theorem 1.7.1 (Hogg et al. 2019): Let X be a continuous random variable with pdf $f_X(x)$ and support $\mathcal{S}_X$. Let $Y = g(X)$ where $g(x)$ is a one-to-one *[monotonic] differentiable* function on the support of X , $\mathcal{S}_X$. Denote the inverse of g by $x = g^{-1}(y)$ and $dx/dy = d[g^{-1}(y)]/dy$. Then, the pdf of Y is given by

$$f_Y(y) = f_X(g^{-1}(y)) \left| \frac{dx}{dy} \right|, \text{ for } y \in \mathcal{S}_Y$$

where the support of Y is the set $\mathcal{S}_Y = \{y = g(x) : x \in \mathcal{S}_X\}$

Getting density of the ratio

Therefore, with $X = u$ and $g^{-1}(y) = \ln(1 + y)$

$$f_{\frac{e}{M}}(y) = f_u(\ln(1 + y)) \left| \frac{1}{1 + y} \right| \quad (27)$$

Informal proof

If $U = g(R)$ for a strictly monotonic differentiable g , then

$$f_R(r) = f_U(g(r))|g'(r)|$$

Proof (via CDFs). Take g strictly increasing (your case). Then

$$F_R(r) = P(R \leq r) = P(g(R) \leq g(r)) = P(U \leq g(r)) = F_U(g(r)),$$

using that g increasing preserves the inequality direction.

Differentiate both sides in r :

$$f_R(r) = \frac{d}{dr}F_U(g(r)) = f_U(g(r))g'(r).$$

If g were decreasing instead, the inequality flips ($P(U \geq g(r)) = 1 - F_U(g(r))$), and differentiating introduces a minus sign that exactly cancels the sign of $g'(r)$, hence $|g'(r)|$ of the general formula.

Everything goes through unchanged conditional on $Z = z$ (just work with $F_{R|Z}(\cdot | z)$, $F_{u|Z}(\cdot | z)$ throughout).

Discussion

Deconvolution estimates the unconditional f_u because of how we set it up

- ▶ From f_u , we can learn the probability of a firm of overreporting u ,
 $P(U < u) = \int_0^u f_u(t)dt$, or
 - ▶ the expected overreporting $\mu_u = \int u f_u(u)du$
- ▶ This is the unconditional probability, the same for all firms regardless of their characteristics
- ▶ We might be interested in the *probability of overreporting conditional* on the firm size

What if we are interested in the conditional density?

$$\begin{aligned} f_{V|Z}(V | Z) &= \frac{f_{V,Z}(V, Z)}{f_Z(Z)} \\ &= \int f_{V|u,Z}(V | u, Z) f_{u|Z}(u | Z) du \\ &= \int f_{\varepsilon}(V - u) f_{u|Z}(u | Z) du \end{aligned}$$

using

- ▶ $f_{V|u,Z}(V | u, Z) = f_{\varepsilon|u,Z}(V - u | u, Z) - V = u + \varepsilon$ conditional on u
- ▶ $\varepsilon \perp (u, Z)$ **jointly**, so $f_{\varepsilon|u,Z} = f_{\varepsilon}$

with $u \not\perp Z$ (otherwise conditioning on Z would be pointless).

Estimating the Conditional distribution

- ▶ **Discrete/bucketed** Z (e.g. quintiles of K_{it} or L_{it}): the same 1-D deconvolution separately within each bucket.
 - ▶ We've already looked into $E[u|Z]$
- ▶ **Continuous** Z : nonparametric deconvolution with a continuous covariate
 - ▶ Literature: $f_{u|V}(u|V)$ (Wang and Ye 2015), $E[Z|u]$ (Fan and Truong 1993), and $f_{Z|u}$ (Huang and Zhou 2020)
 - ▶ $f_{u|Z}$: feasible using characteristic functions and kernels
- ▶ Ideally, what I want is $f_{e|\omega}$, but that's what f_{ψ} is, the object I will obtain with the structural approach

Conditional Deconvolution — Backup

What About the Full Distribution?

- ▶ So far: $E[u_{ijt} \mid Z_{ijt}]$, and only at 10 decile bins of Z
- ▶ Natural next question: the whole conditional density $f_{u|Z}(u \mid z)$, for Z continuous
 - ▶ Is the size gradient in the *level* of evasion, its *spread*, or its *shape*?

Aside: How Deconvolution Works

We never observe u_{ijt} directly — only $V_{ijt} = u_{ijt} + \varepsilon_{ijt}$, a “blurred” version of it

- ▶ A histogram of V mixes the true shape of u with the noise ε — the density of a sum of independent things is the *convolution* of their densities, $f_V = f_u * f_\varepsilon$
- ▶ Convolutions are hard to undo directly, but they become simple *multiplication* after a Fourier transform (the characteristic function):

$$\phi_V(t) = \phi_u(t) \phi_\varepsilon(t) \quad \implies \quad \phi_u(t) = \frac{\phi_V(t)}{\phi_\varepsilon(t)}$$

- ▶ Estimate ϕ_V from the data, divide by the already-known ϕ_ε , transform back $\rightarrow \hat{f}_u$
- ▶ This is exactly the (unconditional) trick behind the first-stage results (Carroll and Hall 1988; Stefanski and Carroll 1990)

Conditional Version: Add a Local Window

Same trick, done separately for each value of Z :

- ▶ Instead of using *all* firms to build one blurred distribution, use only firms with Z_i close to z — a moving local window, weighted by a kernel K
- ▶ Un-blur *that* local distribution the same way as before

$$\hat{\phi}_{V|Z=z}(t) = \frac{\sum_i K\left(\frac{Z_i-z}{h}\right) e^{itV_i}}{\sum_i K\left(\frac{Z_i-z}{h}\right)} \implies \hat{f}_{u|Z=z}(u) = \frac{1}{2\pi} \int e^{-itu} \frac{\hat{\phi}_{V|Z=z}(t)}{\phi_\varepsilon(t)} \omega(t) dt$$

- ▶ $\omega(t)$: a damping weight, standard in any deconvolution, to keep the division numerically stable
- ▶ Valid because $\varepsilon \perp (u, Z)$: within any Z -window, the noise is still independent of u — the same assumption used throughout this section

Is This Standard?

- ▶ (Fan and Truong 1993), already cited on the “is it feasible?” slide, addresses the *mirror-image* problem: a noisy covariate and a clean outcome (see also (Huang and Zhou 2020) for the conditional-density version of that same direction)
- ▶ Ours runs the other way — clean Z , noisy u — which is why the conditional *mean* never needed correcting: $\varepsilon \perp Z \Rightarrow E[V \mid Z] = E[u \mid Z]$
- ▶ Only the *density* needs deconvolving, and the tool is just the classical deconvolution kernel (Carroll and Hall 1988; Stefanski and Carroll 1990), applied inside a local window — a natural combination, not (to our knowledge) a named result in the literature

Do We Need the Full Density?

- ▶ The conditional mean already carries the headline story: evasion rises then falls with size — going further only pays off if it answers a question the mean can't
- ▶ The one moment that would matter: the conditional **variance** $\text{Var}(u \mid Z = z)$
 - ▶ Does “large firms don't evade” mean (a) *every* large firm individually evades a little — tight distribution near zero, like the corporations — or (b) a mix of some firms still evading heavily and others near zero that cancel out in the mean?
 - ▶ (a) vs. (b) is a real audit-targeting question: under (a), size alone is a good targeting rule at the top; under (b) it isn't — some large firms are still hiding substantial evasion behind the average
- ▶ If pushed further: the full nonparametric conditional deconvolution (previous slides) is the general method; cumulant/method-of-moments matching is a cheap over-identification check on the variance estimate, not a substitute for it

Appendix

Appendix

B) Value of Reported Ghost Purchases Over Firm Total Purchases, by Firm Size

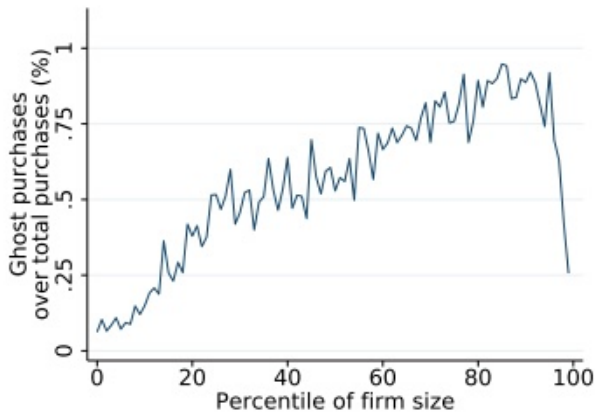


Figure 1: Figure 1B from Carrillo et al. (2022) ↵

Robust Check for Size Heterogeneity

$$\mathcal{V}_{ijt} = \sum_{z \in Z} \sum_{h \in 1,2} \mu_{z,h} x_{z,ijt}^h + \eta_{ijt} \quad (28)$$

Table 1: Coefficient estimates . 95% Bootstrap CI with 250 replications. Stars denote Bootstrap p-value significance at (***) 0.001, (**) 0.05, and (*) 0.1 level.

	IV: Lagged Revenue	OLS: Lagged Revenue
Revenue	1.8012*** [1.5744, 2.0744]	
Revenue ²	-0.7594*** [-1.0265, -0.5149]	
Revenue _{t-1}		1.456*** [1.274, 1.6735]
Revenue _{t-1} ²		-0.5586*** [-0.784, -0.339]
Labour	-0.7149*** [-0.9244, -0.5631]	-0.4281*** [-0.6214, -0.2994]
Labour ²	0.1051 [-0.0561, 0.3035]	-0.09 [-0.2306, 0.1051]
Capital	-0.1124 [-0.277, 0.085]	-0.022 [-0.1776, 0.1737]
Capital ²	-0.2332** [-0.4548, -0.0406]	-0.2997*** [-0.5255, -0.1106]
Exports	-0.1127 [-0.7196, 0.3843]	-0.2759 [-0.8917, 0.192]
Exports ²	0.066 [-0.483, 0.7506]	0.2503 [-0.2569, 0.9386]
Imports	-1.1048*** [-1.4765, -0.7346]	-1.1253*** [-1.5047, -0.7651]
Imports ²	1.3084*** [0.8792, 1.7581]	1.3469*** [0.9235, 1.8055]
Const.	-0.1309*** [-0.1662, -0.0985]	-0.1158*** [-0.1492, -0.0826]

Table 2: Coefficient estimates. Percentile of size of different variables. 95% Bootstrap CI with 250 replications. Stars denote Bootstrap p-value significance at (***) 0.001, (**) 0.05, and (*) 0.1 level.

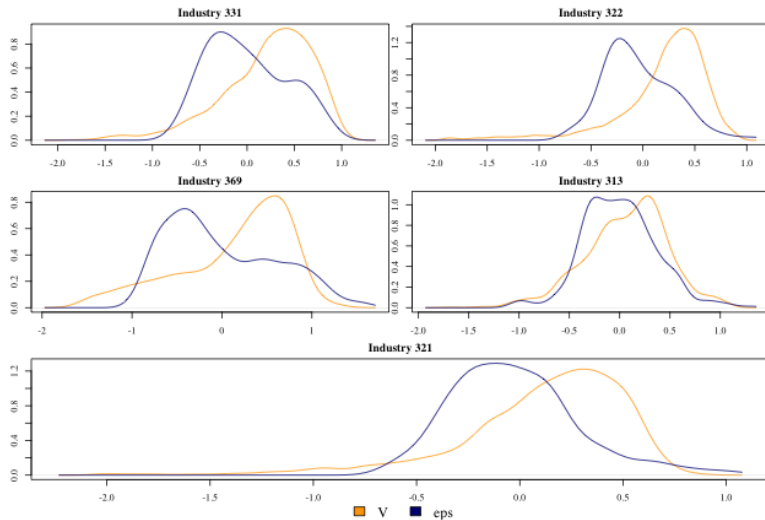
	IV: Lagged Revenue	OLS: Lagged Revenue
Revenue	1.1491*** [1.0863, 1.2497]	
Revenue _{t-1}		0.9948*** [0.9362, 1.0738]
Labour	-0.6258*** [-0.7001, -0.5599]	-0.5044*** [-0.5691, -0.4402]
Capital	-0.3082*** [-0.3679, -0.2278]	-0.2737*** [-0.3309, -0.1943]
Exports	-0.1004 [-0.5756, 0.283]	-0.2292 [-0.7139, 0.1385]
Imports	-0.7292*** [-0.9883, -0.5182]	-0.7386*** [-1.0034, -0.5334]

Why the Negative Sign for Some Small Firms?

- ▶ Smaller firms are more likely to be Proprietorships and less formal
- ▶ Proprietorships report their industrial activity as part of their individual income taxes. They need to keep track of their personal expenses and their industrial activity expenses
- ▶ Because they are more likely to be less formal than larger firms, they may lack a rigorous expense report system
- ▶ Thus, they might be consistently underreporting their expenses

Densities of $\mathcal{V}$ and ε

V and Epsilon Densities by Industry



Empirical Exercises using $\mathcal{V}$

Empirical exercises remain valid as long as changes in $\mathcal{V}$ (tax evasion proxy) are not explained by changes in M (or ε) but only by e or e/M

- ▶ Changes in the proxy are explained by changes to incentives to adjust overreporting not firms adjusting their optimal choice of true inputs.
- ▶ Fiscal Reform of 1983:
 - ▶ Alternative argument, changes in true inputs explain the increase in the overreporting share. How? Suppose e is unchanged. This implies that after the policy reform unincorporated firms decrease their level of inputs by either an increase in relative prices or productivity shocks. Productivity shocks are unrelated to tax rates, to this is unlikely. According to the FOC, taxes do not show in the optimal choice of firms. Besides, we see an increase after the reform, while the reform made only a one-time change. So this alternative argument does not hold.
 - ▶ Our argument is more transparent: an increase in the sales tax rate lead to an increase in the overreporting share because a higher tax rate increases the marginal benefit of overreporting.

Empirical Exercises using $\mathcal{V}$ (Continued)

- ▶ Tax Evasion heterogeneity along firm size:
 - ▶ Alternative argument, correlation between true inputs and size explain why the overreporting share increases along the size distribution and drops at the top. How? fix e . CD could not be capturing the relationship between labour (or capital) and materials. A non-linear relationship could explain the pattern in the data.
 - ▶ However, the pattern is consistent with evidence from other papers with administrative data. As a robustness check, I also use the lag and double lag labour and capital and the pattern persist.

Maximum Simulated Likelihood (MSL)

Solving for M and e

NEW From Equation 19, I can *algebraically solve* for true inputs as function of observed variables and an the error term,

► $M_{it} = M_{it}^* \mathbf{e}^{-\mathcal{V}_{it} - \varepsilon_{it}}.$

Therefore,

$$\begin{aligned} e_{it} &= M_{it}^* - M_{it} \\ &= M_{it}^* (1 - \mathbf{e}^{-\mathcal{V}_{it} - \varepsilon_{it}}) \end{aligned}$$

- ε_{it} is not observed firm-by-firm, only its distribution f_{ε}
- not a literal point recovery of e_{it} , but the map that could be evaluated for some draw of $\varepsilon^{(s)} \sim \hat{f}_{\varepsilon}$

Recast in terms of one unobservable

I can do the same and solve for ω_{it} from Equation 23

For each value of ε everything else is pinned down:

$$e_{it}(\varepsilon) = M_{it}^* (1 - \mathbf{e}^{-\mathcal{V}_{it} - \varepsilon}) \quad (29)$$

$$\omega_{it}(\varepsilon) = \tilde{\mathcal{W}}_{it} - (1 - \beta)\varepsilon \quad (30)$$

$$\psi_{it}(\varepsilon) = h(e_{it}(\varepsilon)) - \delta_0 + \delta_1 \omega_{it}(\varepsilon) - \delta_2 \omega_{it}(\varepsilon)^2. \quad (31)$$

MSL

- ▶ We could estimate f_ψ and $\theta = \{\lambda, \delta_0, \delta_1, \delta_2\}$ by Maximum Simulated Likelihood (MSL)
- ▶ After the recast, one unobservable remains: ε_{it} , with known distribution f_ε (from stage 1),
- ▶ we also know the distribution of ω
- ▶ I would need a *density transformation* to get from the primitives' density $f_{\psi, \omega, \varepsilon}(\psi, \omega, \varepsilon)$ to the density of the observables $f_{\mathcal{V}, \tilde{\mathcal{W}}}(\mathcal{V}, \tilde{\mathcal{W}})$
 - ▶ accounting for the nonlinear map from ψ to $\mathcal{V}$ (Theorem 7.5.3, next slide) *and*
 - ▶ integrating out ε <!--

$$f_{\mathcal{V}_{it}, \tilde{\mathcal{W}}_{it}}(\mathcal{V}_{it}, \tilde{\mathcal{W}}_{it}) = \int f_\psi(\psi(\varepsilon_{it})) f_\omega(\omega(\varepsilon_{it})) f_\varepsilon(\varepsilon_{it}) |\det J\Phi^{-1}(\varepsilon)| d\varepsilon_{it}$$

The object to estimate is f_ψ and $\theta = (\lambda, \delta_0, \delta_1, \delta_2)$, since f_ω and f_ε are already known →

Theorem 7.5.3 Change of Variables in R^n Judd (1998)

Assume that $\Phi : X \subset R^n \rightarrow R^n$ is C^1 and that there is a C^1 inverse map $\Phi^{-1} : Y \subset R^n \rightarrow X$ where $Y = \Phi(X)$ and Y and X are open. If f is integrable on X , then

$$\int_Y f(y)dy = \int_X f(\Phi(x)) |\det J(\Phi)(x)| dx,$$

where $J(\Phi)(x)$ is the Jacobian of Φ at x .

Corollary: Density transformation

Let $f_X(x)$ be the joint pdf of X , B any region in the support of Y , and $A = \Phi^{-1}(B)$

For $Y = \Phi(X)$: $P(X \in A) = P(\Phi(X) \in \Phi(A)) = P(Y \in B)$. Apply the theorem with $\Psi \equiv \Phi^{-1}$ in place of Φ (Φ^{-1} is C^1 with C^1 inverse Φ) to $g = f_X \cdot \mathbf{1}_A$:

$$P(X \in A) = \int_A f_X(x) dx = \int_B f_X(\Phi^{-1}(y)) |\det J\Phi^{-1}(y)| dy.$$

Since B is arbitrary, and using $J\Phi^{-1}(y) = [J\Phi(x)]^{-1}$ (inverse function theorem) at $x = \Phi^{-1}(y)$, then, the joint pdf of Y is:

$$f_Y(y) = \frac{f_X(x)}{|\det J\Phi(x)|} \bigg|_{x=\Phi^{-1}(y)} \quad (32)$$

Target Density

To find the correct target density of observables, note that, given (K_{it}, L_{it}) , the variables $(\mathcal{V}_{it}, \tilde{\mathcal{W}}_{it})$ form an affine map in the (Y_{it}, M_{it}^*) -space,

$$\mathcal{V}_{it} = m_{it}^* - y_{it} + c_1 \quad (19 \text{ rearranged})$$

$$\tilde{\mathcal{W}}_{it} = y_{it} - \beta(m_{it}^* - \mathcal{V}_{it}) - \alpha_K k_{it} - \alpha_L l_{it} \quad (\text{equivalent to 23})$$

where $c_1 \equiv \ln \left(\frac{\rho_t}{P_t \beta} \right)$

Substituting the first equation into the second cancels the m_{it}^* terms exactly:

$$\tilde{\mathcal{W}}_{it} = (1 - \beta) y_{it} + \beta c_1 - \alpha_K k_{it} - \alpha_L l_{it} \quad (33)$$

Φ is a linear bijection in (y_{it}, m_{it}^*) -space

$$\Phi : (y_{it}, m_{it}^*) \mapsto (\mathcal{V}_{it}, \tilde{\mathcal{W}}_{it})$$

$$\begin{pmatrix} \mathcal{V} \\ \tilde{\mathcal{W}} \end{pmatrix} = \underbrace{\begin{pmatrix} -1 & 1 \\ 1 - \beta & 0 \end{pmatrix}}_A \begin{pmatrix} y \\ m^* \end{pmatrix} + \begin{pmatrix} c_1 \\ \beta c_1 - \alpha_K k - \alpha_L l \end{pmatrix}.$$

Since $\det A = -(1 - \beta) \neq 0$ ($0 < \beta < 1$) $\implies \Phi$ is an affine bijection

- ▶ $\implies (\mathcal{V}_{it}, \tilde{\mathcal{W}}_{it})$ contain the same information as (Y_{it}, M_{it}^*)
- ▶ Therefore, the correct target density $f_{\mathcal{V}, \tilde{\mathcal{W}}|K, L}$
- ▶ **No need** to include M^* in the joint density or condition on $(\mathcal{V}$ contains all information of $M^*)$

Conditioning on ω_{it-1} instead of (K_{it}, L_{it})

Due to the Markov process of productivity, $\omega_{it} = g(\omega_{it-1}) + \eta_{it}$,
 $\eta_{it} \perp\!\!\!\perp (\omega_{it-1}, k_{it}, l_{it})$,

$$f_{\omega_{it}|\omega_{it-1}, K_{it}, L_{it}}(\omega_{it}) = f_{\omega_{it}|\omega_{it-1}}(\omega_{it}) = f_{\eta}(\omega_{it} - g(\omega_{it-1}))$$

Conditional on ω_{it-1} , the only randomness left in ω_{it} is η_{it} , which doesn't depend on (K_{it}, L_{it})

► ω_{it-1} is a **sufficient statistic**: once it's known, (K_{it}, L_{it}) adds nothing.

Setting Φ in the $(\psi_{it}, \omega_{it}, \varepsilon_{it})$ -space

$\Phi : (\psi_{it}, \omega_{it}, \varepsilon_{it}) \mapsto (\mathcal{V}_{it}, \tilde{\mathcal{W}}_{it}, a)$, padding with the auxiliary $a \equiv \varepsilon_{it}$ so
 $\Phi : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ matches the theorem's dimension requirement:

Define $G(\psi_{it}, \omega_{it}) \equiv u(e^*(\psi_{it}, \omega_{it}))$,

- ▶ where $u(e) = \ln((M_{it} + e)/M_{it})$, holding M_{it} fixed as e varies $(M \perp e \mid \omega)$, and
- ▶ $e^*(\psi_{it}, \omega_{it}) = h^{-1}(\delta_0 - \delta_1 \omega_{it} + \delta_2 \omega_{it}^2 + \psi_{it})$ inverts the FOC for e_{it} ,

$$\mathcal{V} = G(\psi, \omega) - \varepsilon, \quad \tilde{\mathcal{W}} = \omega + (1 - \beta)\varepsilon, \quad a = \varepsilon$$

- ▶ Like (K_{it}, L_{it}) before, ω_{it-1} does not enter the map

Conditional density

Let $J_i(\varepsilon_{it}) = |\det J\Phi^{-1}|$, then,

$$f_{\mathcal{V}_{it}, \tilde{\mathcal{W}}_{it} | \omega_{it-1}}(\mathcal{V}_{it}, \tilde{\mathcal{W}}_{it}) = \int f_{\psi}(\psi(\varepsilon_{it})) J_i(\varepsilon_{it}) f_{\eta}\left(\omega(\varepsilon_{it}) - g(\omega_{it-1})\right) f_{\varepsilon}(\varepsilon_{it}) d\varepsilon_{it}. \quad (34)$$

Assuming $\psi_{it}, \varepsilon_{it} \perp (K_{it}, L_{it}) \mid \omega_{it-1}$

Since ω_{it-1} isn't observed, use $\tilde{\mathcal{W}}_{it-1}$

$\tilde{\mathcal{W}}_{it-1}$ is firm- i *data*

- ▶ ε_{it-1} unobserved, but known density f_ε (stationarity) and
- ▶ already assumed independent $\varepsilon_{it-1} \perp \varepsilon_{it}, \eta_{it}$ in productivity's AR(1) estimation
- ▶ extending $\varepsilon_{it-1} \perp \psi_{it}$ (additional assumption)
- ▶ therefore, we can marginalize ε_{it-1} with an ordinary integral,

Likelihood contribution of a firm

$$\begin{aligned} f_{\mathcal{V}_{it}, \tilde{\mathcal{W}}_{it} | \tilde{\mathcal{W}}_{it-1}}(\mathcal{V}_{it}, \tilde{\mathcal{W}}_{it} | \tilde{\mathcal{W}}_{it-1}) = \\ \iint f_{\psi}(\psi(\varepsilon_{it})) J_i(\varepsilon_{it}) f_{\eta}\left(\omega(\varepsilon_{it}) - g(\tilde{\mathcal{W}}_{it-1} - (1 - \beta)\varepsilon_{it-1})\right) \\ \times f_{\varepsilon}(\varepsilon_{it}) f_{\varepsilon}(\varepsilon_{it-1}) d\varepsilon_{it} d\varepsilon_{it-1}. \end{aligned} \tag{35}$$

Recovering f_η

Define the residual used in the AR(1) productivity process,

$$\zeta_{it} \equiv \mathcal{W}_{it} - \alpha_K k_{it} - \alpha_L l_{it} - \gamma_0 - \gamma_1 (\mathcal{W}_{it-1} - \alpha_K k_{it-1} - \alpha_L l_{it-1}) = (1 - \beta) \varepsilon_{it} - \gamma_1$$

a sum of three mutually independent variables, two with known densities, f_ε .

Fourier deconvolution recovers the third:

$$\phi_\eta(t) = \frac{\phi_\zeta(t)}{\phi_\varepsilon((1 - \beta)t) \phi_\varepsilon(-\gamma_1(1 - \beta)t)} \longrightarrow f_\eta,$$

The Jacobian matrix

$$J\Phi = \frac{\partial(\mathcal{V}, \tilde{\mathcal{W}}, a)}{\partial(\psi, \omega, \varepsilon)} = \begin{pmatrix} G_\psi & G_\omega & -1 \\ 0 & 1 & \beta \\ 0 & 0 & 1 \end{pmatrix}$$

where $G_\psi \equiv \partial G / \partial \psi$ and $G_\omega \equiv \partial G / \partial \omega$

Determinant

Expand along column 1 (only G_ψ is nonzero there):

$$\det J\Phi = G_\psi \cdot \det \begin{pmatrix} 1 & 1 & -\beta \\ 0 & 1 & 1 \end{pmatrix} = G_\psi \cdot 1 = G_\psi$$

- ▶ G_ω never enters because every cofactor multiplying it comes from a zero in column 1.
- ▶ Mechanically: $\tilde{\mathcal{W}} = \omega + (1 - \beta)\varepsilon$ has **unit coefficient on ω** , a flat translation, so it contributes no curvature.
- ▶ A Jacobian only appears where the map is genuinely curved: the $\psi \rightarrow \mathcal{V}$ link through G , not $\omega \rightarrow \tilde{\mathcal{W}}$.

Unpacking G_ψ

$G = u \circ e^*$, so by the chain rule $G_\psi = u'(e^*) \cdot e_\psi^*$.

► $u'(e)$: from $u(e) = \ln(M_{it} + e) - \ln M_{it}$, with M_{it} fixed,

$$u'(e) = \frac{1}{M_{it} + e}$$

evaluated at $e = e^*$, $M_{it} + e^* = M_{it}^*$, $\implies u'(e) = 1/M_{it}^*$

► e_ψ^* : differentiate the FOC identity $\psi = h(e^*(\psi, \omega)) - \delta_0 + \delta_1\omega - \delta_2\omega^2$ w.r.t. ψ (implicit function theorem, valid since $h' \neq 0$):

$$1 = h'(e^*) \cdot e_\psi^* \implies e_\psi^* = \frac{1}{h'(e^*)}$$

Combine:

$$G_\psi = \frac{1}{M_{it}^* h'(e)}$$

Assembling $J_i(\varepsilon)$

From Equation 32, the multiplying factor is $1/|\det J\Phi| = 1/|G_\psi|$:

$$J_i(\varepsilon) \equiv \frac{1}{|G_\psi|} = |h'(e)| M_{it}^*$$

Every piece is traceable: - $|h'(e)|$ from inverting the FOC ($\psi \rightarrow e$), - M_{it}^* from the log-ratio definition ($e \rightarrow \mathcal{V}$, holding the *true* M_{it} fixed) - they multiply because $G = u \circ e^*$ is a **composition**, so the chain rule stacks them.

Marginalizing out the auxiliary

Applying the density formula gives the joint density of $(\mathcal{V}, \tilde{\mathcal{W}}, a)$, but only $(\mathcal{V}, \tilde{\mathcal{W}})$ are observed;

- ▶ Integrating a out (an ordinary single integral, not part of the theorem) gives the likelihood contribution of firm i :

$$f_{\mathcal{V}, \tilde{\mathcal{W}}}(\mathcal{V}_{it}, \tilde{\mathcal{W}}_{it}) = \int f_{\psi}(\psi(\varepsilon)) J_i(\varepsilon) f_{\omega}(\omega(\varepsilon)) f_{\varepsilon}(\varepsilon) d\varepsilon \quad (36)$$

A 3-for-3 change of variables that ends in a 1-D integral:

- ▶ the theorem handles the square transformation and hands back a Jacobian;
- ▶ marginalizing the unobserved auxiliary coordinate is the separate, ordinary step left over once the theorem's job is done.

Simulated log-likelihood

$$\mathcal{L}(\theta, f_\psi) = \sum_{i \in \text{Unincorp}} \ln \left(\int f_\psi(\psi(\varepsilon)) J_i(\varepsilon) f_\omega(\omega(\varepsilon)) f_\varepsilon(\varepsilon) d\varepsilon \right) \quad (37)$$

Simulating the likelihood

Approximate Equation 36 by **Monte Carlo**

$$\widehat{\mathcal{L}}_i(\theta) = \frac{1}{S} \sum_{s=1}^S f_{\psi}(\psi(\varepsilon^{(s)})) J_i(\varepsilon^{(s)}) f_{\omega}(\omega(\varepsilon^{(s)})), \quad \varepsilon^{(s)} \stackrel{iid}{\sim} \hat{f}_{\varepsilon} \quad (38)$$

- ▶ Draws $\varepsilon^{(s)}$ come from *exactly* the density $\hat{f}_{\varepsilon}(\varepsilon) d\varepsilon$ multiplying the integrand — this is plain Monte Carlo integration, unbiased for **any** $S \geq 1$
- ▶ Alternatively, use **deterministic quadrature** to integrate Equation 36

Why this works despite ε entering e nonlinearly

- ▶ $e(\varepsilon) = M_{it}^* (1 - \mathbf{e}^{-\mathcal{V}_{it} - \varepsilon})$ is nonlinear in ε
- ▶ But Equation 38 averages the **density value** $f_{\psi}(\psi(\varepsilon^{(s)})) J_i(\varepsilon^{(s)}) f_{\omega}(\omega(\varepsilon^{(s)}))$ at each draw
 - ▶ never averages $\psi(\varepsilon^{(s)})$ *by itself*
- ▶ Monte Carlo integration of a (possibly wild) function against its own sampling density is unbiased regardless of the function's shape
 - ▶ nonlinearity only breaks things if the reweighting is skipped
- ▶ Every simulated $\varepsilon^{(s)} \neq$ the firm's true ε_{it} , almost surely
 - ▶ but draws implying an implausible (ψ, ω) pair get down-weighted by $f_{\psi} \cdot J_i \cdot f_{\omega}$; only draws near the truth carry weight,
 - ▶ so the weighted average reconstructs the correct posterior-weighted integral even though no single draw is the truth

Convergence — a known theorem

Lee (1992): if the simulator of a likelihood contribution is (i) unbiased and continuous in θ , and (ii) built from draws i.i.d. across firms and replicates, then under standard M-estimation regularity (compactness, identification, dominance) the simulated-MLE is consistent and $\sqrt{N}$ -equivalent to the infeasible exact MLE, **provided** $\sqrt{N}/S \rightarrow 0$ as $N, S \rightarrow \infty$ (fixed S leaves an $O(1/S)$ bias, since $\ln(\text{average}) \neq \text{average of } \ln$)

- ▶ Unbiasedness (i): shown above, holds by construction
- ▶ Continuity (ii): h , q , and $J_i(\varepsilon)$ are smooth throughout the interior domain $e \in (0, \frac{1}{2\lambda})$
- ▶ Draws i.i.d. across i, t, s , independent of θ : by construction

An even cleaner argument: quadrature

- ▶ The preferred implementation is **deterministic quadrature**, not Monte Carlo simulation, for this 1-D integral
- ▶ Quadrature avoids simulation noise because it is deterministic and can be driven arbitrarily small by adding nodes
- ▶ The $\sqrt{N}/S \rightarrow 0$ condition above is then moot: with negligible quadrature error, ordinary MLE consistency applies directly

Caveat: this is a two-step estimator

- ▶ β and f_ε are **first-stage estimates**, not the truth
- ▶ Still consistent under standard conditions, but naive stage-2 standard errors ignore stage-1 estimation uncertainty
- ▶ Use the Murphy and Topel (1985) two-step correction, or bootstrap both stages jointly

References

- Allingham, Michael G., and Agnar Sandmo. 1972. "Income Tax Evasion: A Theoretical Analysis." *Journal of Public Economics* 1 (November): 323–38. [https://doi.org/10.1016/0047-2727\(72\)90010-2](https://doi.org/10.1016/0047-2727(72)90010-2).
- Carrillo, Paul, Dave Donaldson, Dina Pomeranz, and Monica Singhal. 2022. *Ghosting the Tax Authority: Fake Firms and Tax Fraud*. NBER. <http://www.nber.org/papers/w30242>.
- Carroll, Raymond J., and Peter Hall. 1988. "Optimal Rates of Convergence for Deconvolving a Density." *Journal of the American Statistical Association* 83 (404): 1184–86. <https://doi.org/10.1080/01621459.1988.10478718>.
- Fan, Jianqing, and Young K. Truong. 1993. "Nonparametric Regression with Errors in Variables." *The Annals of Statistics* 21 (4): 1900–1925. <https://doi.org/10.1214/aos/1176349402>.